

Unit 2- Analytical Questions

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Course Code / Name: CSA-5104 / Cryptography and Network Security

1) → Plaintext: 0123456789ABCDEF

Key: 133457799BBCDFFI

DES Encryption Process:

Step:	Description:
Initial Permutation	The 64-bit plaintext is rearranged using DES IP table
Key Scheduling	The 64-bit key is reduced to 56 bits by removing parity bits.
Round 1 Feistel Operation	The plaintext is divided into Left (L_0) and Right (R_0) halves. R_0 is expanded to 48 bits, XORed with K_1 , passed through S-boxes and P-boxes, the XORed with L_0 to produce new right half.
Round 2-16	The same Feistel Opr. is repeated using round keys K_2 to K_{16}

2) Plaintext: 11112222 33334444
Key: AABB 091827 36CCDD

Step:	Description:
IP	The plaintext undergoes DES IP
Key Scheduling	16, 48 bit round keys are generated from the given key using PC-1, left shifts, and PC-2.
Feristel Rounds	The data block passes through 16 Feristel rounds consisting of Expansion, XOR, S-box, P-box.
Final P	The inverse permutation is applied to obtain ciphertext.
Result	The ciphertext is obtained after final DES permutation.

3) Plaintext: 1³2456 ABCD 132536

Key: AABB CCDD EE FF 0011

Encryption Steps:

- Apply IP.
- Generate 16 round keys from given key.
- Perform 16 Feristel rounds.
- Swap the left and right halves.
- Apply FP.
- Obtain encrypted 64 bit ciphertext.

Plaintext: FED CBA987 6543210

Key: DF1571C 947D9E859

Step	Description
IP	Rearranges plaintext bits accordingly to the IP table
Key Schedule	Generates 16 round keys using PC-1, circular left shifts, EPC2
Round Operations	Each round performs Expansion $\rightarrow$ XOR $\rightarrow$ S-box sub $\rightarrow$ P-box P $\rightarrow$ XOR with left half
FP	Final inverse permutation produces the ciphertext.

5) Plaintext: 02468 ACEE CA86420

Key: DE 329232 EAD0D73

DES Encryption:

1. Apply Initial Permutation
2. Gen 16 subkeys
3. Perform 16 Feistel rounds
4. Swap halves.
5. Apply FP
6. Output the encrypted ciphertext.

6) Plaintext: 0123456789 ABCDEF

Key: FED CBA9876543210 0123456789 ABCDEF

Step	Description
Init.	Init. the P-array and four S-boxes using Blowfish constants
Key Expansion	XOR the user key with P-array and repeatedly encrypt an

	all-zero block to update the P-array and S-boxes.
Feistel Rounds	Split the plaintext into two 32-bit halves and perform 16 Feistel rounds using the Blowfish F-function.
Final Swap	Swap the left and right halves after the 16 th round.
Final XOR	XOR with the last two P-array entries to produce the ciphertext.

7) Plaintext: 123456789ABCDEF

Key: 00112233445566778899AABBCCDDEEFF

Process:

- * Init P-array and S-boxes.
- * Expand 128-bit Key.
- * Perform 16 Feistel rounds
- * Apply final swap and XOR
- * Produce the encrypted ciphertext

8) Plaintext: FEDCBA9876543210

Key: AABCCDDEEFF00112233445566778899

Steps:

1. Init P-arrays and S-boxes.
2. Perform key expansion.
3. Divide plaintext into two halves.
4. Execute sixteen Feistel rounds.
5. Apply for final swap
6. XOR with the last P-array values.
7. Obtain ciphertext.

and perform

Plaintext: 89ABCDEF01234567

Key: 123456789ABCDEF FEDCBA0987654321

Steps	Description:
P-array usage	The P-array values are used sequentially in each Feistel round
F-function	The right half passes through the F-Function using 4 S-box lookups, modular addition, and XOR opr.
16 Rounds	16 Feistel rounds are executed
Final step	Final XOR and swap produce the ciphertext.

10) Plaintext: ABCDEF01234567890

Key: 0F1E2D3C4B5A 69788796A5B4C3D2E1FD

Step	Description
Init	Init the Blowfish P-array and S-boxes
Key Expansion	Gen the key-dependent P-array and S-box values.
Feistel Rounds	Perform 16 Feistel rounds using the Blowfish F-function & P-array entries
Final XOR	XOR with the final two P-array values.
Ciphertext	The resulting 64-bit encrypted ciphertext is produced.